

Answer Key

Question 1

Fisher equation: $(1 + \text{Real return}) = (1 + \text{Nominal return}) / (1 + \text{Inflation})$

$$\text{Real return} = (1.10 / 1.04) - 1 \approx 5.77\%$$

Question 2

$$\text{Real return} = (1.15 / 1.06) - 1 \approx 8.49\%$$

Question 3

$$\text{Real return} = (1.07 / 1.08) - 1 \approx -0.93\%$$

Even though the investment posted a positive nominal return of 7%, inflation of 8% ran ahead of it. In purchasing-power terms, the investor actually lost ground: the same basket of goods and services that cost \$100 at the start of the year would cost roughly \$108 by year-end, but the investment only grew to about \$107. This is a useful case to notice because a positive nominal return can still mean a real loss whenever inflation exceeds the nominal rate; the sign of the real return isn't guaranteed just because the nominal return is positive.

Question 4

Inflation compounds over time in the same way returns do, so even a modest inflation rate erodes a large share of purchasing power over a long horizon. A portfolio that grows at 8% nominally per year but faces 4% inflation is really only growing at roughly 3.85% in purchasing-power terms, and that gap, applied over 20 or 30 years, produces an enormous difference in real wealth compared to the nominal figure alone. Because retirement portfolios, pension funds, and other long-term goals are ultimately about what money can *buy* in the future rather than its face value, investors and planners should evaluate long-term performance using real returns, not nominal ones. Failing to do so risks overstating how much wealth has actually been built.